https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	r is the independent variable and *f* (frequency of turntable) is the dependent variable **or** vary *r* and measure *f* (frequency of turntable)	**1**
	keep *m* constant	**1**
	Methods of data collection	
	labelled diagram showing power supply connected to motor (two leads) within turntable; circuits must be workable	**1**
	method to change frequency of rotation of the turntable, e.g. adjust output of (variable) power supply or adjust variable resistor	**1**
	increase frequency until the cube moves (relative to the turntable)	**1**
	method to determine period of rotation of the turntable, e.g. stopwatch, light gate attached to a timer/data-logger or stroboscope	**1**
	Method of analysis	
	plots a graph of *f* against 1 / *r* (allow log *f* against log *r*)	**1**
	relationship valid if a straight line produced passing through the origin (for lg *f* vs. lg *r* straight line of gradient of –1)	**1**
	K = gradient $\times 4\pi^2 m$ (for lg *f* vs. lg *r*, $K = 10^{y\text{-intercept}} \times 4\pi^2 m$)	**1**

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Question	Answer	Marks
	Additional detail including safety considerations	**Max. 6**
	D1 use safety screen	
	D2 time at least 10 rotations of turntable or detailed use of stroboscope	
	D3 $f = 1 / T$ for correct determination of period of rotation of turntable	
	D4 repeat experiment for each r and average f	
	D5 use balance to measure mass of cube	
	D6 wait for turntable to rotate steadily before increasing frequency **or** gradual/incremental/slowly increase in frequency	
	D7 use a spirit level to check that turntable is horizontal **or** clean cube/surface	
	D8 use a rule to measure r	
	D9 method to ensure r is measured to the centre of the cube, e.g. put a mark on the cube or align front or back of cube by a set distance	
	D10 method to determine centre of the turntable e.g. measure two or more diameters/maximum distance ideas	

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